

Professional Profile

Applied Artificial Intelligence and Data Engineering undergraduate with research interests in intelligent decision systems, financial machine learning, and enterprise-scale analytics. Through research and industry experience across Hong Kong, the United States, and India, I have developed scalable AI applications, data engineering pipelines, and decision-support systems that bridge machine learning research with real-world business problems. My long-term goal is to contribute to trustworthy and impactful AI through graduate research and industry innovation.

Research Interests

Applied Artificial Intelligence • Decision Intelligence • Financial Machine Learning • Data Engineering
• Enterprise AI • Predictive Analytics • Time Series Forecasting

Education

2023 – 2027 **Bachelor of Technology (B.Tech.)**, *CT University*, Ludhiana, India, CGPA: 9.26 / 10.00
Specialization: Artificial Intelligence & Data Science

Distinctions Academic Excellence Award; Consistent Semester Rank Holder (1st/2nd place across 6 semesters).

Selected Coursework Advanced AI, Machine Learning, Data Mining, Time Series Forecasting, Optimization Techniques,
Computer Vision, Database Systems, Data Structures & Algorithms, Probability & Statistics.

Research Publications

Working Paper Sidhu, M.S. (2026). *Volatility Dynamics of Asian Stock Markets: Evidence from the Hang Seng Index and Nikkei 225*. Zenodo. DOI: 10.5281/zenodo.18180779

Working Paper Sidhu, M.S. (2026). *Generalization Limits in Noisy Time Series Returns*. Zenodo. DOI: 10.5281/zenodo.18180884

Professional Experience

2026 – **Data Engineer Intern**, *LunarTech AI*, Remote, USA

Present Engineered scalable Medallion Architecture pipelines for the ingestion, validation, normalization, and canonicalization of over 100,000 investor records, producing approximately 44,600 high-quality CRM-ready entities.

2026 – **Data Analyst Intern**, *Soulracle Limited*, Hong Kong SAR

Present Conducted product analytics, customer journey mapping, and competitive market analysis across Hong Kong, Taiwan, and Mainland China to support AI product strategy and user experience optimization.

2025 **Software Engineer Intern**, *Byteoski Developers*, Ludhiana, India

Developed AI-powered SaaS platforms using Django, Next.js, PostgreSQL, OpenAI APIs, and Celery, implementing scalable backend services, asynchronous processing pipelines, and business analytics dashboards.

Selected Research & Technical Projects

EcomSWOT Developed an AI-powered SaaS platform that converts Shopify customer review datasets into board-ready SWOT analyses using large language models, sentiment analysis, and automated executive reporting.

SmartExpense AI Designed an intelligent financial assistant that utilizes OCR and machine learning to automate receipt scanning, expense categorization, and expenditure visualization.

- Credit Risk Model Constructed a predictive ML system to estimate loan default probability using supervised learning algorithms on financial feature sets.
- MatchHub Engineered a collaborative networking platform designed to enhance communication and project synergy within university student ecosystems.

Technical Skills

- Programming Python • SQL • JavaScript • Java • C++
- Artificial Intelligence Large Language Models (LLMs) • Machine Learning • Natural Language Processing • Computer Vision • Predictive Analytics
- Data Engineering ETL Pipelines • Medallion Architecture • BigQuery • Entity Resolution • Data Normalization
- Tools & Platforms Django • Next.js • PostgreSQL • Tableau • Git • Celery • REST APIs

Leadership & Volunteering

- 2024 – Pres. **Technical Coordinator**, *TechVerse Club, CTU*
Coordinated technical workshops, hackathons, and peer learning initiatives, contributing to technical activities serving a student community of approximately 800 students.
- Community Volunteer at **Mind HK**; Developer for the 'I Love People' community charitable initiative.

Certifications & Languages

- Certifications IBM Machine Learning, AI, Data Science Foundations, Python for Data Science, Big Data 101, Enterprise Data & AI; British Airways Data Science Virtual Job Simulation.
- Languages Punjabi (Native), English (Fluent), Hindi (Fluent), Cantonese (Beginner), Mandarin (Beginner).